

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276798

Kind Code

A1

Publication Date

September 04, 2025

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TWO-PIECE ENGINE COLLAR

Abstract

An engine collar, aircraft, and method of installing an engine in an aircraft are provided. The engine collar includes a first engine collar piece, including a first portion of a ring, a second engine collar piece, including a second portion of a ring, and a plurality of splice plates joining the first engine collar piece to the second engine collar piece; where each of the first engine collar piece and the second engine collar piece have two engine isolation mounting points and a plurality of attachment points for connecting the engine collar to a plurality of engine supports.

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Family ID: 96880890

Appl. No.: 18/594746

Filed: March 04, 2024

Publication Classification

Int. Cl.: B64D27/40 (20240101); B64C25/04 (20060101); B64F5/10 (20170101)

U.S. Cl.:

CPC B64D27/402 (20240101); B64D27/40 (20240101); B64D27/404 (20240101);
B64D27/406 (20240101); B64D2033/0293 (20130101); F02C7/20 (20130101);
F05D2230/60 (20130101)

Background/Summary

TECHNICAL FIELD

[0001] The present invention relates generally to an improved engine collar design in an aircraft using a turboprop engine, and more specifically to an engine collar that is capable of supporting 45 degree) (45° engine isolators while still maintaining ideal drop-in installation/removal of a turboprop engine.

BACKGROUND

[0002] The General Electric™ turboprop engine for the Beechcraft™ Denali™ presented a novel issue of having four 45° isolators located along the engine circumference. The 45° orientation reduced engine vibration, improved engine performance and reduced the engine box structural weight. However, as legacy turboprop engines typically had four 30° isolators with a single collar piece, a new design was necessary. Utilizing previous designs with a single collar piece would require both the structural trusses and collar to be installed on the new turboprop engine prior to installation in the aircraft, reducing efficiency, and making engine drop-in/removal much more complicated and costly. Previously used bent-tube engine collars further exhibited a high scrap rate due to a lack of precision with aligning multiple attachment location. Accordingly, a new design was needed that still enabled an easy drop-in engine installation, reduced scrap rate, and provided necessary precision to align the collar splice, (4) isolators and support truss tubes.

SUMMARY

[0003] In accordance with a preferred embodiment of the present invention, an improved two-piece machined engine collar comprises as follows.

[0004] In a first embodiment, an engine collar includes: a first engine collar piece, including a first portion of a ring, and having a back portion, where the back portion is flat on a first side and has, on a second side opposite the first side, a raised outer sidewall along an outer circumference, a raised inner sidewall along an inner circumference, and a plurality of connecting walls spaced apart from each other running from the raised outer sidewall to the raised inner sidewall, and where the connecting walls are substantially perpendicular to a tangent of the raised inner sidewall and raised outer sidewalls along a major axis of the connecting walls; a second engine collar piece, including a second portion of a ring, and having a back portion, where the back portion is flat on a first side and has, on a second side opposite the first side, a raised outer sidewall along an outer circumference, a raised inner sidewall along an inner circumference, and a plurality of connecting walls spaced apart from each other running from the raised outer sidewall to the raised inner sidewall, where the connecting walls are substantially perpendicular to the tangent of the raised inner sidewall and raised outer sidewalls along the major axis of the connecting walls, and where the first engine collar piece including the first portion of the ring and the second engine collar piece including the second portion of the ring together include a complete ring; and a plurality of splice plates joining the first engine collar piece to the second engine collar piece; where each of the first engine collar piece and the second engine collar piece have two engine isolation mounting points and a plurality of attachment points for connecting the engine collar to a plurality of engine supports.

[0005] In a second embodiment, an aircraft includes an engine, four engine isolators, a plurality of engine supports, and an engine collar, where the engine collar includes: a first engine collar piece, having a back portion flat on a first side, and, on a second side opposite the first side, having a raised outer sidewall along an outer circumference of the first engine collar piece, a raised inner sidewall along an inner circumference of the first engine collar piece, and a plurality of connecting walls spaced apart from each other running a length between the raised outer sidewall and the raised inner sidewall; a second engine collar piece, having a back portion flat on a first side, and, on a second side opposite the first side, having a raised outer sidewall along an outer circumference of the second engine collar piece, a raised inner sidewall along an inner circumference of the second engine collar piece, and a plurality of connecting walls spaced apart from each other

running a length between the raised outer sidewall and the raised inner sidewall; and a plurality of splice plates joining the first engine collar piece to the second engine collar piece at a narrow portion of the engine, where the engine is wider forward of the narrow point of the engine and wider aft of the narrow point of the engine; where the first engine collar piece and the second engine collar piece together form a ring around the engine; and where the engine collar is connected to the engine on a first side of the engine collar by the four engine isolators, and where the engine collar is connected to the aircraft on a second side of the engine collar through the plurality of engine supports.

[0006] In a third embodiment a method of installing an engine in an aircraft includes: attaching four engine isolators to the engine at mounting points spaced around a circumference of the engine, where the engine is narrower at the engine mounting points than at least of a portion of the engine forward of the engine mounting points, and a portion of the engine aft of the engine mounting points; attaching a first engine collar piece to two of the four engine isolators using fasteners protruding through two engine isolation mounting points protruding inward from an inner circumference of the first engine collar piece; attaching a second engine collar piece to the remaining two of the four engine isolators using fasteners protruding through two engine isolation mounting points protruding inward from an inner circumference of the second engine collar piece; joining the first engine collar piece to the second engine collar piece using a plurality of splice plates and fasteners extending through the first engine collar piece or the second engine collar piece to form an engine collar; connecting the engine using the engine collar by dropping the engine into position on a plurality of first supports in the aircraft; fastening at least one of the first engine collar piece or the second engine collar piece to the plurality of first supports; installing a plurality of second supports in the aircraft; and fastening the plurality of second supports to the engine collar.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0008] FIGS. **1** through **4** are alternative views of a two-piece engine collar design according to some embodiments;

[0009] FIG. **5** is a partial view of the splice plates joining the two pieces of the engine collar according to some embodiments;

[0010] FIG. **6** is a view of the two-piece engine collar installed on an aircraft engine support system according to some embodiments;

[0011] FIG. **7** is a view of the two-piece engine collar and turboprop engine installed on an aircraft engine support system according to some embodiments; and

[0012] FIG. **8** is a method for installing an engine according to some embodiments.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0013] Illustrative embodiments of the system of the present application are described below. In the interest of clarity, not all features of an actual implementation are described in this specification. It will of course be appreciated that in the development of any such actual embodiment, numerous implementation-specific decisions must be made to achieve the developer's specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure.

[0014] In the specification, reference may be made to the spatial relationships between various

components and to the spatial orientation of various aspects of components as the devices are depicted in the attached drawings. However, as will be recognized by those skilled in the art after a complete reading of the present application, the devices, members, apparatuses, etc. described herein may be positioned in any desired orientation. Thus, the use of terms to describe a spatial relationship between various components or to describe the spatial orientation of aspects of such components should be understood to describe a relative relationship between the components or a spatial orientation of aspects of such components, respectively, as the device described herein may be oriented in any desired direction.

[0015] Embodiments of the presented principles are directed toward an improved machined two-piece engine collar design used in, for example, an airframe utilizing a turboprop engine. The system provides high-quality parts that can be customized, resulting in tighter tolerance, improved manufacturing reliability, reduced scrap rates, reduced vibration transfer from a turboprop engine to the airframe, improved engine performance, and reduced the engine box structural weight, while still maintaining ideal drop-in/removal capability for ease of construction, maintenance, and replacement.

[0016] Traditional turboprop engine isolators were located at 30 degrees) (30° offsets on either side of a centerline diameter (i.e., at 60 degree) (60° spacing between a first and second engine isolator and a third and fourth engine isolator with 120 degree) (120° spacing between the second and third engine isolator and the fourth and first engine isolator when viewed in sequence around the engine circumference). Some new turboprop engine designs have presented a novel issue of having four engine isolators located at 45 degrees) (45° along the engine circumference (i.e., the spacing between each engine isolator being 90 degrees)) (90°. The 45° orientation results in reduced engine vibration, improved engine performance, and reduced engine box structural weight. However, the new engine isolator spacing presents a challenge in how to drive the engine load from the engine isolators back to the airframe while still maintaining ideal drop-in installation/removability that is present in other engine support structures. A two-piece machined collar design was able to achieve this design goal while also providing other advantages.

[0017] FIG. 1 shows a first view of an assembled two-piece engine machined collar design according to some embodiments. The engine collar **100** has an upper collar piece **110**, a lower collar piece **120** and one or more splice plates **130** are used to bind the upper collar piece **110** to the lower collar piece. Each of the upper collar piece **110** and the lower collar piece **120** has an outer sidewall **140**, an inner sidewall **150**, a solid back plate **160**, and a number of connecting walls **170** between the outer sidewall **140** and the inner sidewall **150**.

[0018] In some embodiments, the upper collar piece **110** and the lower collar piece **120** may each comprise approximately 180 degrees) (180° of the total circumference of the of the engine collar **100**. For the purposes of this disclosure “approximately 180 degrees” here means within a few degrees of 180 degrees) (180° with sufficient space to allow for mechanical and fabrication tolerances between the upper collar piece **110** and the lower collar piece **120**. In other embodiments, one collar piece may comprise more than 180 degrees) (180° of the total circumference of the collar **100** such that the other collar piece comprises less than 180 degrees) (180° of the total circumference of the collar **100**.

[0019] In some embodiments, the upper collar piece **110** and the lower collar piece **120** may be fabricated from 4340 steel with a tensile yield strength of 180 to 200 kilopounds per square inch (KSI). In some embodiments the upper collar piece **110** and the lower collar piece **120** may conform to the AMS6414 standard as published by SAE International.

[0020] In some embodiments the upper collar piece **110** and the lower collar piece **120** may be fabricated using a negative removal fabrication process such as a milling machine, a computer numerical control (CNC) router, or the like. In some embodiments, the thickness of the solid back plate **160** of the upper collar piece **110** and the lower collar piece **120** may be 0.10 inches $\pm 10\%$.

[0021] Engine isolation mounting points **180** are located at approximately 45 degrees) (45° from a

centerline as shown by angles A1 through A4. For the purposes of this disclosure, approximately 45 degrees means ± 2.5 degrees) (2.5°). In some embodiments, the engine isolation mounting points **180** may be located on tabs extending in towards a center point of the engine collar **100** from the inner sidewall **150**.

[0022] FIG. 2 illustrates an engine collar **100** with the splice plates **130** separated from the engine collar **100** body according to some embodiments. The upper collar piece **110** and the lower collar piece **120** are held together on each side of the engine collar **100** by a back splice plate **210**, a front splice plate **220**, an inner wall inner splice plate **230**, and inner wall outer splice plate **240**, and outer wall inner splice plate **250**, and an outer wall outer splice plate **260** (collectively, “splice plates **130**”). In some embodiments, the back splice plate **210** and the front splice plate **220** are fastened together on either side of the solid back plate **160** portion of the upper collar piece **110** and the lower collar piece **120** using fasteners **270** that extend through the back splice plate **210**, the front splice plate **220**, and the solid back plate **160** portion of either the upper collar piece **110** or the lower collar piece **120**. Similarly, the inner wall inner splice plate **230** and the inner wall outer splice plate **240** are held together by fasteners **270** that extend through the inner wall inner splice plate **230** and the inner wall outer splice plate **240**, as well as the inner sidewall **150** of either the upper collar piece **110** or the lower collar piece **120**. Further, the outer wall inner splice plate **250** and the outer wall outer splice plate **260** are held together by fasteners **270** that extend through the outer wall inner splice plate **250**, the outer wall outer splice plate **260**, and the outer sidewall **140** of either the upper collar piece **110** or the lower collar piece **120**. In some embodiments, the fasteners are nuts and bolts with washers on either side of the fastening surfaces. In other embodiments, the fasteners may include screws, rivets, or the like.

[0023] In some embodiments, the splice plates **130** may each be fabricated from 17-7PH stainless steel. In some embodiments the splice plates **130** may conform to the AMS 5528, AMS 5529, and/or AMS 5644 standards as published by SAE International. In other embodiments, the splice plates may be fabricated from another suitable material such as a metal, alloy, polymer, composite, ceramic, or the like, or a combination of materials.

[0024] FIG. 3 illustrates an alternate view of the engine collar **100** according to some embodiments. As shown, the back splice plate **210** may further be seen connecting the upper collar piece **110** and the lower collar piece **120**. In some embodiments, the engine collar **100** may further include nose wheel well fittings **310** and **320** attached to the lower collar piece **120**. The nose wheel well fittings **310** and **320** may be connected to the engine collar via fasteners such as bolts, and a landing gear assembly (not shown here) may be connected to protruding tabs of the nose wheel well fittings **310** and **320** using, for example, pins, bolts, or the like.

[0025] In some embodiments, the distance D.sub.1 of the upper collar piece **110** and the lower collar piece **120** from the back face to the top of the outer sidewall **140** and/or inner sidewall **150** may be 1.0 inches $\pm 10\%$. In some embodiments, the engine isolation mounting points **180** may protrude further than the sidewalls of the upper collar piece **110** and the lower collar piece **120** such that a distance D.sub.2 from the back face to the top of engine isolation mounting points **180** may be 2.0 inches $\pm 10\%$.

[0026] FIG. 4 illustrates an alternative view of the engine collar **100** according to some embodiments. The upper collar piece **110** and the lower collar piece **120** are joined to each other utilizing the splice plates **130**. In some embodiments, an inner diameter D.sub.inner of the engine collar **100** may be 25.75 inches $\pm 10\%$. In some embodiments an outer diameter D.sub.outer of the engine collar **100** may be 20.50 inches $\pm 10\%$. This results in a width W.sub.1 of the engine collar **100** being approximately 5.25 inches $\pm 10\%$.

[0027] In some embodiments, the engine collar **100** includes mounting holes **330** for attaching the engine collar **100** to engine support trusses (not shown here).

[0028] FIG. 5 is partial view of the engine collar **100** showing the splice plates **130** according to some embodiments. In some embodiments, the outer sidewall **140** and inner sidewall **150** have a

thickness of 0.15 inches $\pm 10\%$. In some embodiments, the connecting walls **170** have thicknesses between 0.05 inches and 0.10 inches. In some embodiments, the sidewall interface **510** between the inner sidewall **150** or the outer sidewall **140** and the connecting walls **170** may have a rounded shape. Further, the backplane interface **520** between inner sidewall **150**/outer sidewall **140**/connecting walls **170** and the solid back plate **160** may be rounded as well. In other embodiments, an angular or squared sidewall interface and/or backplane interface may be utilized. [0029] In some embodiments, the splice plates **130** each have a thickness **T1** of 0.08 inches $\pm 10\%$. In some embodiments spacing D.sub.4 between fasteners **270** for the inner wall inner splice plate **230** and the inner wall outer splice plate **240** is between 1.14 inches to 1.30 inches. In some embodiments spacing D.sub.5 between fasteners **270** for the outer wall inner splice plate **250** and the outer wall outer splice plate **260** is between 0.96 inches to 1.16 inches. In some embodiments, spacing D.sub.6 between rows of fasteners **270** for the back splice plate **210** (not shown) and the front splice plate **220** is between 1.25 inches to 1.35 inches. Spacing D.sub.7 between rows of fasteners for the back splice plate **210** (not shown) and the front splice plate **220** is 1.0 inch $\pm 10\%$. In some embodiments, where a bolt would be located on the joint between the upper collar piece **110** and the lower collar piece **120**, a larger distance D.sub.8 between bolts may be selectively utilized.

[0030] FIG. **6** illustrates a potential embodiment of the engine collar **100** installed in an aircraft frame. Engine supports **620** and **625** attach to the engine collar **100** at points around the engine collar **100** on one side of each engine support **620** and **625**, and to a forward pressure bulkhead **630** of the airframe **750**. The engine supports **620** and **625**, support the weight of the engine (not shown), and transfer thrust produced by the engine to the airframe **750**. Engine isolators **610** connect the engine collar **100** to the engine (not shown). In some embodiments, the engine isolators **610** may include a dampening system to limit vibration transfer from the engine (not shown) to the airframe **750**.

[0031] FIG. **7** shows an embodiment of the engine collar **100** attachment to a turboprop engine **710**. In some embodiments, a nose landing gear wheel well **720** may be attached to the engine collar **100** at the nose wheel well fittings **310** and **320** (see FIG. **3**) using secondary supports **730**. Accordingly, the engine collar **100** may provide additional load bearing capability for the nose landing gear wheel well **720**, in conjunction with engine supports **620** and **625**. As can be seen in the figure, the engine collar **100** is connected to mounting points on the turboprop engine **710** through engine isolators **610**.

[0032] In some embodiments, to install a turboprop engine in an aircraft using the above described engine collar is shown in FIG. **8**. Installation may comprise the following steps. The order of steps may be varied in some embodiments.

[0033] **S101** comprises attaching four engine isolators **610** to the turboprop engine **710** at mounting points spaced around a circumference of the turboprop engine **710**. In **S102**, the upper collar piece **110** is attached to two of the four engine isolators **610** using two engine isolation mounting points **180** protruding inward from an inner circumference of the upper collar piece **110**. In **S103**, the lower collar piece **120** may then be attached to the remaining two of the four engine isolators **610** using two engine isolation mounting points **18** protruding inward from an inner circumference of the lower collar piece **120**. In some embodiments the engine isolators **610** may be attached to the upper collar piece **110** and lower collar piece **120** using fasteners such as bolts, screws, pins, rivets, or the like.

[0034] In **S104**, the upper collar piece **110** and lower collar piece **120** may then be joined together around the turboprop engine **710**. In some embodiments the upper collar piece **110** and lower collar piece **120** may be joined using a plurality of splice plates **130**. In some embodiments, the upper collar piece **110** and the lower collar piece **120** may be held together on each side of the engine collar **100** by a back splice plate **210**, a front splice plate **220**, an inner wall inner splice plate **230**, and inner wall outer splice plate **240**, and outer wall inner splice plate **250**, and an outer wall outer

splice plate **260** (collectively, “splice plates **130**”). In some embodiments, the back splice plate **210** and the front splice plate **220** are fastened together on either side of the solid back plate **160** portion of the upper collar piece **110** and the lower collar piece **120** using fasteners **270** that extend through the back splice plate **210**, the front splice plate **220**, and the solid back plate **160** portion of either the upper collar piece **110** or the lower collar piece **120**. Similarly, the inner wall inner splice plate **230** and the inner wall outer splice plate **240** are held together by fasteners **270** that extend through the inner wall inner splice plate **230** and the inner wall outer splice plate **240**, as well as the inner sidewall **150** of either the upper collar piece **110** or the lower collar piece **120**. Further, the outer wall inner splice plate **250** and the outer wall outer splice plate **260** are held together by fasteners **270** that extend through the outer wall inner splice plate **250**, the outer wall outer splice plate **260**, and the outer sidewall **140** of either the upper collar piece **110** or the lower collar piece **120**. In some embodiments, the fasteners are nuts and bolts with washers on either side of the fastening surfaces. In other embodiments, the fasteners may include screws, rivets, or the like.

[0035] In some embodiments, the joined engine collar **100** encircles the turboprop engine **710** at a narrow point towards the middle of the turboprop engine **710**. In some embodiments, at least of a portion of the turboprop engine **710** forward of the engine mounting points, and a portion of the turboprop engine **710** aft of the engine mounting points is wider than the opening in the engine collar **100**. In some embodiments, **S104** may be performed before either or both of **S102** and/or **S103** described above.

[0036] **S015** comprises dropping the combined turboprop engine **710** and engine collar **100** into position in the airframe **750**. In some embodiments, a first plurality of engine supports **625** are not yet installed in the airframe **750**. Accordingly, in **S106**, the engine collar **100** is attached to a second plurality of engine supports **620** installed in the airframe **750** prior to dropping the turboprop engine **710** into position in **S105**. This permits installation of the turboprop engine **710** with the engine collar **100** already installed, and with the engine isolators **610** in approximately 90 degree spacing from each other, and spaced about 45 degrees from the engine collar **100** join points.

[0037] In some embodiments, the engine supports **620** attach to the engine collar **100** at points around the engine collar **100** on one side of each engine support **620**, and to a forward pressure bulkhead **630** of the airframe **750** on an opposite side of each engine support **620**. In **S107**, the first plurality of engine supports **625** may then be installed between the forward pressure bulkhead **630** of the airframe **750**. In **S108**, the first plurality of engine supports **625** may be fastened to the engine collar. The engine supports **620** and **625** may be fastened to the engine collar **100** using bolts, screws, pins, rivets, or the like.

[0038] In some embodiments, optional **S109** may include attaching a nose landing gear wheel well **720** to the lower collar piece **120** by secondary supports **730**, where the secondary supports **730** provide structural support to the nose landing gear wheel well **720** through the engine collar **100**, and the engine supports **620** and **625**.

[0039] Due to the various protrusions and the irregular shape of the turboprop engine **710**, merely removing the engine isolators **610** from the engine collar **100** would not facilitate drop-in installation or removability while still maintaining proper support to drive the engine load from the engine isolators **610** back to the airframe **750**. However, by utilizing a two-piece engine collar design as described above, tight tolerances may be maintained to properly support and align a turboprop engine **710** within an airframe, resulting in reduced engine vibration, improved engine performance, and reduced engine box structural weight. Further, by removing fasteners **740** to disconnect the engine collar **100** from the engine supports **620**, the turboprop engine **710** (with the engine collar **100** and engine isolators still attached) may be easily removed from the airframe **750** for maintenance, inspection, replacement, or the like, with minimal disassembly required.

Additionally, a two-piece engine collar design exhibits a lower scrap rate in manufacturing than conventional methods, improving repeatability and cost.

[0040] In one embodiment, an engine collar includes a first engine collar piece, including a first

portion of a ring, and having a back portion, where the back portion is flat on a first side and has, on a second side opposite the first side, a raised outer sidewall along an outer circumference, a raised inner sidewall along an inner circumference, and a plurality of connecting walls spaced apart from each other running from the raised outer sidewall to the raised inner sidewall, and where the connecting walls are substantially perpendicular to a tangent of the raised inner sidewall and raised outer sidewalls along a major axis of the connecting walls; a second engine collar piece, including a second portion of a ring, and having a back portion, where the back portion is flat on a first side and has, on a second side opposite the first side, a raised outer sidewall along an outer circumference, a raised inner sidewall along an inner circumference, and a plurality of connecting walls spaced apart from each other running from the raised outer sidewall to the raised inner sidewall, where the connecting walls are substantially perpendicular to the tangent of the raised inner sidewall and raised outer sidewalls along the major axis of the connecting walls, and where the first engine collar piece including the first portion of the ring and the second engine collar piece including the second portion of the ring together include a complete ring; and a plurality of splice plates joining the first engine collar piece to the second engine collar piece; where each of the first engine collar piece and the second engine collar piece have two engine isolation mounting points and a plurality of attachment points for connecting the engine collar to a plurality of engine supports.

[0041] In some embodiments of the engine collar, each engine isolation mounting point is 87.5 degrees to 92.5 degrees from a closest next engine isolation mounting point around a circumference of the engine collar. In some embodiments, each engine isolation mounting point is disposed in a tab extending in from the raised inner sidewall of the first engine collar piece or the second engine collar piece, where each tab is shaped as a rounded protrusion inward from the raised inner sidewall along a major plane of the engine collar, and where each engine isolation mounting point includes a raised hollow cylinder at least partially disposed outside a circumference of the raised inner sidewall and extending further than the raised inner sidewall in a direction perpendicular to the major plane of the engine collar. In some embodiments, the plurality of splice plates includes at least two of each of a back splice plate, a front splice plate, an inner wall inner splice plate, and inner wall outer splice plate, and outer wall inner splice plate, and an outer wall outer splice plate. In some embodiments, the first portion of the ring included by the first engine collar piece is 178 degrees to 182 degrees of a circle, and the second portion of the ring included by the second engine collar piece completes a remainder of the circle. In some embodiments, an engine isolation mounting point is located at approximately 45 degrees from each end of each of the first engine collar piece and the second engine collar piece. In some embodiments, the second engine collar piece further includes two landing gear assembly connection points. In some embodiments, the first engine collar piece and second engine collar piece are formed from 4340 steel. In some embodiments, each splice plate of the plurality of splice plates are formed from 17-7PH stainless steel.

[0042] In another embodiment, an aircraft includes an engine, four engine isolators, a plurality of engine supports, and an engine collar, where the engine collar includes: a first engine collar piece, having a back portion flat on a first side, and, on a second side opposite the first side, having a raised outer sidewall along an outer circumference of the first engine collar piece, a raised inner sidewall along an inner circumference of the first engine collar piece, and a plurality of connecting walls spaced apart from each other running a length between the raised outer sidewall and the raised inner sidewall; a second engine collar piece, having a back portion flat on a first side, and, on a second side opposite the first side, having a raised outer sidewall along an outer circumference of the second engine collar piece, a raised inner sidewall along an inner circumference of the second engine collar piece, and a plurality of connecting walls spaced apart from each other running a length between the raised outer sidewall and the raised inner sidewall; and a plurality of splice plates joining the first engine collar piece to the second engine collar piece at a narrow

portion of the engine, where the engine is wider forward of the narrow point of the engine and wider aft of the narrow point of the engine; where the first engine collar piece and the second engine collar piece together form a ring around the engine; and where the engine collar is connected to the engine on a first side of the engine collar by the four engine isolators, and where the engine collar is connected to the aircraft on a second side of the engine collar through the plurality of engine supports.

[0043] In some embodiment of the aircraft, each engine isolation mounting point is 87.5 degrees to 92.5 degrees from a closest next engine isolation mounting point around a circumference of the engine collar. In some embodiment of the aircraft, each engine isolation mounting point is located in a tab extending in from the raised inner sidewall of the first engine collar piece or the second engine collar piece, where each tab is shaped as a rounded protrusion inward from the raised inner sidewall along a major plane of the engine collar, where each engine isolation mounting point includes a raised hollow cylinder at least partially located outside a circumference of the raised inner sidewall and extending further than the raised inner sidewall in a direction perpendicular to the major plane of the engine collar, where an engine isolator at least partially overlaps each engine isolation mounting point when installed, and where a fastener extending through each engine isolation mounting point secures each engine isolator to the engine collar. In some embodiment of the aircraft, the plurality of splice plates includes at least two of each of a back splice plate, a front splice plate, an inner wall inner splice plate, and inner wall outer splice plate, and outer wall inner splice plate, and an outer wall outer splice plate. In some embodiment of the aircraft, the second engine collar piece further includes two landing gear assembly connection points that attach to a forward landing gear assembly through secondary supports. In some embodiment of the aircraft, the first engine collar piece and second engine collar piece are formed from 4340 steel. In some embodiment of the aircraft, the plurality of splice plates are formed from 17-7PH stainless steel.

[0044] In a further embodiment a method of installing an engine in an aircraft includes: attaching four engine isolators to the engine at mounting points spaced around a circumference of the engine, where the engine is narrower at the engine mounting points than at least of a portion of the engine forward of the engine mounting points, and a portion of the engine aft of the engine mounting points; attaching a first engine collar piece to two of the four engine isolators using fasteners protruding through two engine isolation mounting points protruding inward from an inner circumference of the first engine collar piece; attaching a second engine collar piece to the remaining two of the four engine isolators using fasteners protruding through two engine isolation mounting points protruding inward from an inner circumference of the second engine collar piece; joining the first engine collar piece to the second engine collar piece using a plurality of splice plates and fasteners extending through the first engine collar piece or the second engine collar piece to form an engine collar; connecting the engine using the engine collar by dropping the engine into position on a plurality of first supports in the aircraft; fastening at least one of the first engine collar piece or the second engine collar piece to the plurality of first supports; installing a plurality of second supports in the aircraft; and fastening the plurality of second supports to the engine collar.

[0045] In some embodiments method of installing an engine in an aircraft further includes attaching a forward landing gear assembly to the second engine collar piece by a third set of supports, where the third set of supports provide structural support to the forward landing gear assembly through the engine collar, the first supports, and the second supports. In some embodiments, the first engine collar piece and second engine collar piece are formed from 4340 steel, and where each engine isolation mounting point is 87.5 degrees to 92.5 degrees from a closest next engine isolation mounting point around a circumference of the engine collar. In some embodiments, the plurality of splice plates are formed from 17-7PH stainless steel and includes at least two of each of a back splice plate, a front splice plate, an inner wall inner splice plate, and inner wall outer splice plate, and outer wall inner splice plate, and an outer wall outer splice plate.

[0046] While this invention has been described with reference to illustrative embodiments, this

description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

Claims

1. An engine collar, comprising: a first engine collar piece, comprising a first portion of a ring, and having a back portion, wherein the back portion is flat on a first side and has, on a second side opposite the first side, a raised outer sidewall along an outer circumference, a raised inner sidewall along an inner circumference, and a plurality of connecting walls spaced apart from each other running from the raised outer sidewall to the raised inner sidewall, wherein the connecting walls are substantially perpendicular to a tangent of the raised inner sidewall and raised outer sidewalls along a major axis of the connecting walls; a second engine collar piece, comprising a second portion of the ring, and having a back portion, wherein the back portion is flat on a first side and has, on a second side opposite the first side, a raised outer sidewall along an outer circumference, a raised inner sidewall along an inner circumference, and a plurality of connecting walls spaced apart from each other running from the raised outer sidewall to the raised inner sidewall, wherein the connecting walls are substantially perpendicular to the tangent of the raised inner sidewall and raised outer sidewalls along the major axis of the connecting walls, and wherein the first engine collar piece comprising the first portion of the ring and the second engine collar piece comprising the second portion of the ring together comprise a complete ring; and a plurality of splice plates joining the first engine collar piece to the second engine collar piece; wherein each of the first engine collar piece and the second engine collar piece have two engine isolation mounting points comprising tabs protruding inward from a circumference of the raised inner sidewall, and a plurality of attachment points for connecting the engine collar to a plurality of engine supports.
2. The engine collar of claim 1, wherein each engine isolation mounting point is 87.5 degrees to 92.5 degrees from a closest next engine isolation mounting point around a circumference of the engine collar.
3. The engine collar of claim 1, wherein each tab is shaped as a rounded protrusion inward from the raised inner sidewall along a major plane of the engine collar, and wherein each engine isolation mounting point comprises a raised hollow cylinder at least partially disposed outside a circumference of the raised inner sidewall and extending further than the raised inner sidewall in a direction perpendicular to the major plane of the engine collar.
4. The engine collar of claim 1, wherein the plurality of splice plates comprises at least two of each of a back splice plate, a front splice plate, an inner wall inner splice plate, and inner wall outer splice plate, and outer wall inner splice plate, and an outer wall outer splice plate.
5. The engine collar of claim 1, wherein the first portion of the ring comprised by the first engine collar piece is 178 degrees to 182 degrees of a circle, and wherein the second portion of the ring comprised by the second engine collar piece completes a remainder of the circle.
6. The engine collar of claim 5, wherein an engine isolation mounting point is located at approximately 45 degrees from each end of each of the first engine collar piece and the second engine collar piece.
7. The engine collar of claim 1, wherein the second engine collar piece further includes two landing gear assembly connection points.
8. The engine collar of claim 1, wherein the first engine collar piece and second engine collar piece are formed from 4340 steel.
9. The engine collar of claim 1, wherein each splice plate of the plurality of splice plates are formed from 17-7PH stainless steel.
10. An aircraft comprising: an engine; four engine isolators; a plurality of engine supports; and an

engine collar, comprising: a first engine collar piece, having a back portion flat on a first side, and, on a second side opposite the first side, having a raised outer sidewall along an outer circumference of the first engine collar piece, a raised inner sidewall along an inner circumference of the first engine collar piece, and a plurality of connecting walls spaced apart from each other running a length between the raised outer sidewall and the raised inner sidewall; a second engine collar piece, having a back portion flat on a first side, and, on a second side opposite the first side, having a raised outer sidewall along an outer circumference of the second engine collar piece, a raised inner sidewall along an inner circumference of the second engine collar piece, and a plurality of connecting walls spaced apart from each other running a length between the raised outer sidewall and the raised inner sidewall; and a plurality of splice plates joining the first engine collar piece to the second engine collar piece at a narrow portion of the engine, wherein the engine is wider forward of the narrow point of the engine and wider aft of the narrow point of the engine; wherein the first engine collar piece and the second engine collar piece together form a ring around the engine; and wherein the engine collar is connected to the engine on a first side of the engine collar by the four engine isolators, and wherein the engine collar is connected to the aircraft on a second side of the engine collar through the plurality of engine supports.

11. The aircraft of claim 10, wherein each engine isolation mounting point is 87.5 degrees to 92.5 degrees from a closest next engine isolation mounting point around a circumference of the engine collar.

12. The aircraft of claim 11, wherein each engine isolation mounting point is located in a tab extending in from the raised inner sidewall of the first engine collar piece or the second engine collar piece, wherein each tab is shaped as a rounded protrusion inward from the raised inner sidewall along a major plane of the engine collar, wherein each engine isolation mounting point comprises a raised hollow cylinder at least partially located outside a circumference of the raised inner sidewall and extending further than the raised inner sidewall in a direction perpendicular to the major plane of the engine collar, wherein an engine isolator at least partially overlaps each engine isolation mounting point when installed, and wherein a fastener extending through each engine isolation mounting point secures each engine isolator to the engine collar.

13. The aircraft of claim 10, wherein the plurality of splice plates comprises at least two of each of a back splice plate, a front splice plate, an inner wall inner splice plate, and inner wall outer splice plate, and outer wall inner splice plate, and an outer wall outer splice plate.

14. The aircraft of claim 10, wherein the second engine collar piece further includes two landing gear assembly connection points that attach to a forward landing gear assembly through secondary supports.

15. The aircraft of claim 10, wherein the first engine collar piece and second engine collar piece are formed from 4340 steel.

16. The aircraft of claim 10, wherein the plurality of splice plates are formed from 17-7PH stainless steel.

17. A method of installing an engine in an aircraft, comprising: attaching four engine isolators to the engine at mounting points spaced around a circumference of the engine, wherein the engine is narrower at the engine mounting points than at least a portion of the engine forward of the engine mounting points and a portion of the engine aft of the engine mounting points; attaching a first engine collar piece to two of the four engine isolators using fasteners protruding through two first tabs located at two engine isolation mounting points on the first engine collar piece, each of the two first tabs protruding inward from an inner circumference of the first engine collar piece; attaching a second engine collar piece to the remaining two of the four engine isolators using fasteners protruding through two second tabs located at two engine isolation mounting points on the second engine collar piece, each of the second two tabs protruding inward from an inner circumference of the second engine collar piece; joining the first engine collar piece to the second engine collar piece using a plurality of splice plates and fasteners extending through the first engine collar piece or the

second engine collar piece to form an engine collar; connecting the engine using the engine collar by dropping the engine into position on a plurality of first supports in the aircraft; fastening at least one of the first engine collar piece or the second engine collar piece to the plurality of first supports; installing a plurality of second supports in the aircraft; and fastening the plurality of second supports to the engine collar.

18. The method of installing an engine in an aircraft of claim 17, further comprising: attaching a forward landing gear assembly to the second engine collar piece by a third set of supports, wherein the third set of supports provide structural support to the forward landing gear assembly through the engine collar, the first supports, and the second supports.

19. The method of installing an engine in an aircraft of claim 17, wherein the first engine collar piece and second engine collar piece are formed from 4340 steel, and wherein each engine isolation mounting point is 87.5 degrees to 92.5 degrees from a closest next engine isolation mounting point around a circumference of the engine collar.

20. The method of installing an engine in an aircraft of claim 17, wherein the plurality of splice plates are formed from 17-7PH stainless steel and comprises at least two of each of a back splice plate, a front splice plate, an inner wall inner splice plate, and inner wall outer splice plate, and outer wall inner splice plate, and an outer wall outer splice plate.
